

Question				Marking details	Marks available				Maths	Prac
					AO1	AO2	AO3	Total		
	(d)	(i)		E_k of particles is proportional to Temp (or $E_k = \frac{3}{2}kT$) (1) So particle E_k (or speed, velocity) needs to be high enough to overcome electrostatic (Coulomb) repulsion (1) ...and allow (nuclear) strong force to come into action (1)	1	1 1		3		
		(ii)		$n = \frac{2.2 \times 10^{22}}{75}$ [or $2.9(3) \times 10^{20}$ seen] (1) $2.9 \times 10^{20} \times 120 \times 10^6 \times 0.8 = 2.8 \times 10^{28}$ Fusion possible (1)			2	2	2	
				Question 9 total	6	9	5	20	10	0